

multilayer barrier on both sides. However, researchers have solved this problem with just the ceramic thin e-strate and a thin-film top encapsulation. Plasma-enhanced chemical vapour deposition was used to print a few layers as well.

Minimising the losses

The problem with majority of LEDs is the high amount of losses. The high-efficiency spectrum of LEDs falls at about 50 per cent. So with the best of LEDs we still have a loss of 50 per cent through heat. A new OLED deposition process based on finite-element method has been developed by Korean equipment maker Dawonis. This can be scaled to large-area deposition, enabling efficient evaporation based OLED TV production.

In OLEDs, around 40 per cent of the total emitted light ends up coupling to surface plasmons at the metal cathode. A nanostructured or nanograting cathode structure into their OLED design could theoretically reduce plasmon loss. Using simulation, researchers at Konica Minolta Laboratory, Japan, have designed a simple model representing a metal and emitting layer structure. Optical effects resulting from arbitrary sub wavelength structures were simulated using finite-element model. Conclusions suggest that using a nanostructured cathode could help significantly reduce plasmon loss.

What do we have here? A blue light emitter, perhaps

Recently, a group of Japanese researchers found a family of twisted organic molecules that could emit blue light. Prof. Takuma Yasuda, advanced electronics materials research division, and his team at Kyushu University, Japan, has developed more efficient, cheaper and stable organometallic molecules for use in OLEDs.

They have used twisted pyrimidine-acridan-derived compounds

Something to learn from fireflies

Another team of scientists, led by Ki-Hun Jeong at Korea Advanced Institute of Science and Technology, South Korea, investigated optical properties of a firefly's light-emitting cuticle. The cuticles have tiny structures that increase light extraction by reducing internal reflection. Inspired by these tiny hierarchical structures, researchers replicated the patterns to create an OLED with 60 per cent increase in light extraction and 15 per cent wider angle of illumination.

to produce brilliant blue glow. The tricky bit is that, one part of the molecule is orthogonal to the other. Changing the substituent at pyrimidine unit in these emitters can finely tune their emissive characteristics, thermal properties and energy gaps between the singlet and triplet states, while maintaining frontier molecular orbital levels and, thereby, optimising their optoelectronic properties.

Biomedical application areas

Apart from OLEDs being used for phones and lamps, applications for OLEDs are very vast, ranging from curved OLED displays that are placed on non-flat surfaces to transparent ones embedded in car windshields and window panes.

UK based PolyPhotonix, in collaboration with Centre for Process Innovation, has developed a wearable blanket. It uses printed OLED lighting to administer phototherapy for possible treatment of a number of skin conditions such as acne, psoriasis, eczema and jaundice. The blanket could also be used for wound healing and anti-inflammatory treatments.

Advanced VR headsets could also drive demands for higher-end OLED displays.

Researchers at University of Tokyo, Japan, have developed an ultra-thin and flexible protective layer to create an air-stable OLED display. This is expected to enable creation of electronic skin (e-skin) displays of blood oxygen meters, e-skin heart rate sensors for athletes and many other applications. With more such inventions, wearable OLEDs are going to be very common in the near



Flexible display

future and could find applications in many more fields that we cannot imagine today.

Lighting up the future

In the future, we might forget LCDs for displays, like we forgot the incandescent bulb for lighting after compact fluorescent light came in. Incandescent bulbs were interesting in their own regard. Their light had a natural yellowish tint. Compact fluorescent lights and LEDs produced a whiter shade of light.

Who knows what OLEDs may bring in! Also, until we have a stable blue light source for organics, it is all talks. What we do know is, these potentially consume lesser energy, are aesthetically pleasing and have wider application areas. Do not forget the flexible wristbands or the curved displays in premium smartphones.

With research into TADF materials, we could potentially have a much cheaper lighting solution. Unless we have a stable blue light source, we will end up paying heavily for the touch bar in MacBook Pro 2016. I wonder if the OLED devices would be reasonably priced, once we have the appropriate solutions. Probably not, that could be just wishful thinking. **EFY**